

ANANDHA KRISHNAN P

Embedded Systems & AI Developer | ECE Undergraduate

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TECHNICAL PROFILE

Electronics and Communication Engineering student specializing in embedded systems, IoT architectures, and computer vision pipelines. Built hardware-AI systems using ESP32, MQTT, and Python backends. Developer of S2C, an AI system converting hand-drawn circuits into solvable electrical models using YOLOv8 and Modified Nodal Analysis. CGPA: 9.21.

Research Interests: Embedded AI • Edge Computing • Computer Vision Systems • IoT Security • Real-Time Systems

EDUCATION

B.E. Electronics and Communication Engineering

Mahendra Engineering College, Namakkal, Tamil Nadu

CGPA: 9.21

Expected Graduation: May 2027

TECHNICAL SKILLS

Programming	C, C++, Python (NumPy, OpenCV), JavaScript, SQL
Embedded Systems	ESP32, ESP8266, Arduino (AVR), Embedded C, Microcontrollers, Edge AI
AI / Vision	YOLOv8, OpenCV, Image Morphology, Skeletonization, Computer Vision
Protocols	UART, I2C, SPI, MQTT, HTTP/REST, WebSocket, IoT Systems
Web & Backend	FastAPI, PostgreSQL, React 19, Node.js, Vite, REST APIs, GitHub
Tools	Git, VS Code, Arduino IDE, Node-RED, Streamlit, TinkerCAD, Linux
VLSI / EDA	Digital Logic Design, Data Structures, Algorithms, Verilog (basics), Cadence Virtuoso (learning)

PROJECTS

S2C — Sketch-to-Circuit AI System • [Python](#) · [YOLOv8](#) · [OpenCV](#) · [NumPy](#) · [Streamlit](#) github.com/anandmsak/S2C_VF

- Detection:** Trained YOLOv8 detector on a custom dataset of 1,200+ circuit images; identifies 6 components — resistor, capacitor, inductor, voltage source, transistor, and ground.
- MNA Solver:** Pure-Python Modified Nodal Analysis (NumPy) reconstructs circuit topology and solves DC models in <10 seconds — zero external simulator dependency.
- Explainability:** Auto-generates human-readable KCL equations, conductance matrix, and KVL loops; designed as an explainable AI tool for circuit analysis education.

PRISM-SAFE 2.0 — Crowd Risk Intelligence System • [ESP32-CAM](#) · [PIR/IR Sensors](#) · [MQTT](#) · [Node-RED](#) github.com/anandmsak/PS_2.0

- Hardware:** ESP32-CAM nodes and PIR/IR sensor arrays stream real-time data across up to 16 sensor zones via MQTT to a cloud pipeline.
- Dashboard:** Node-RED heatmap with multi-level alert escalation; designed for deployment at stadiums, railway stations, and high-density public venues.
- Outcome:** Pitched as a 13-slide technical deck with sensor architecture diagrams and risk flow charts.

R2V — Right to Vote: Digital Voting Platform • [FastAPI](#) · [PostgreSQL](#) · [React 19](#) · [JWT](#) github.com/anandmsak/r2v/tree/a2

- Security:** Hash-chained audit log ensures tamper detection — any vote modification breaks the chain and is immediately flagged.
- Stack:** FastAPI + PostgreSQL backend with JWT auth and invite-key admin portal; React 19 + Vite frontend, JSONB metadata audit trails.
- Scale:** Dual portals (Student / Admin), register-number voter validation, real-time election management for college-level deployment.

Smart Waste Segregation & Monitoring System • [Embedded C](#) · [AI Classification](#) · [IoT Sensors](#) github.com/anandmsak/SIH_25014_Project

- Selected for Smart India Hackathon 2025** (college level) — sensor-driven AI waste classification with real-time monitoring alerts across multiple waste categories.

Smart Eye — Real-Time Crowd Monitoring • [ESP32-CAM](#) · [MQTT](#) · [Node-RED](#)

- 1st Prize, College Project Expo** — ESP32-CAM crowd monitoring node with MQTT data pipeline and planned edge-AI density classification using onboard inference.

ACHIEVEMENTS

- 1st Prize — Project Expo** — Smart Eye, Mahendra Engineering College
- 2nd Prize — Paper Presentation** — SCEINSEA 2K25, Kongu Engineering College
- SIH 2025 — College Level Selection** — Smart Waste Segregation & Monitoring System
- AI & Sustainability Virtual Internship** — Applied ML for green-technology domain

CERTIFICATIONS

- Embedded Systems for Beginners — NIEIT (National Institute of Electronics and Information Technology)
- IoT and Embedded Systems Workshop — Mahendra Engineering College
- Generative AI and Explainable AI (XAI) Workshop
- Ai & sustainability Virtual Internship - AICTE